人机共解

预测模型向可解释耦合智能的范式转移

Human-AI Teaming

A Paradigm Shift from Predictive Models to Explainable Coupled Intelligence

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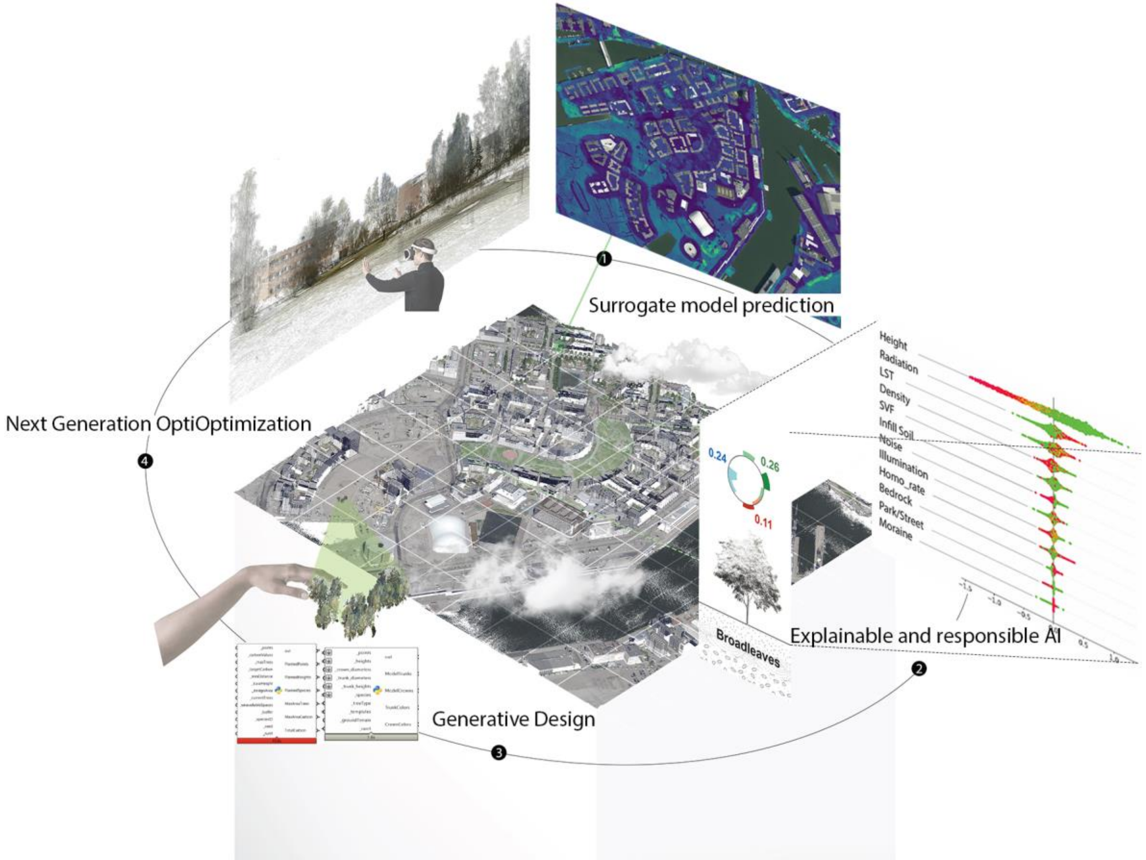


Pia

Chenhao

Yichao

Chaowen

What We Will Explore

Schedule

Design Optimization Needs a New Paradigm

通过引入人工智能与代理模型，本研究将探讨三个关键的前沿方向：

Through adopting AI and surrogate modeling, three key research frontiers will be examined:

可解释与负责任的人工智能
1
Explainable and Responsible AI

下一代优化方法
2
Next Generation Optimization

生成式设计
3
Generative Design

方法论 Methodology

End-to-End Pipeline

1 多维特征提取

Feature Extraction

2 人工智能体训练

Surrogate Model Training

3 分析模块封装

Module Encapsulation

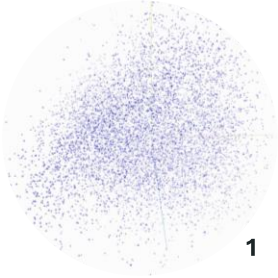
4 可解释智能耦合

XAI Coupling

5 基于场景的空间推测

Scenario Inference

案例 Examples



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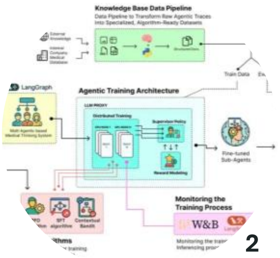

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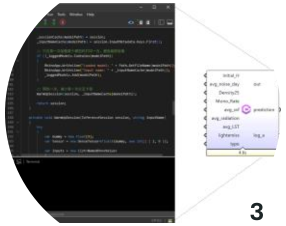

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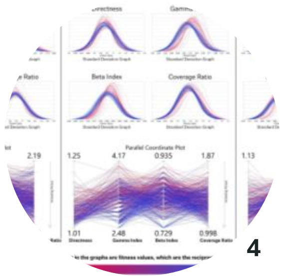

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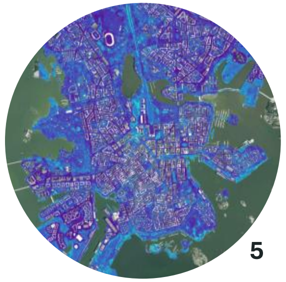

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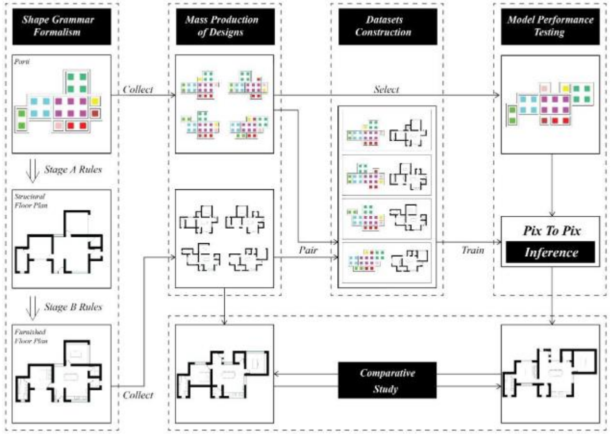
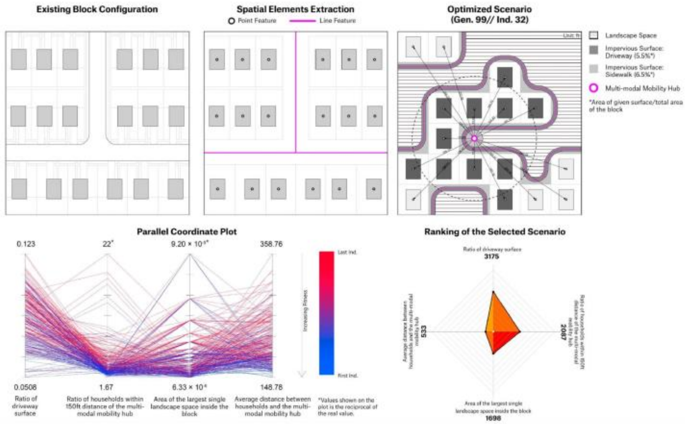
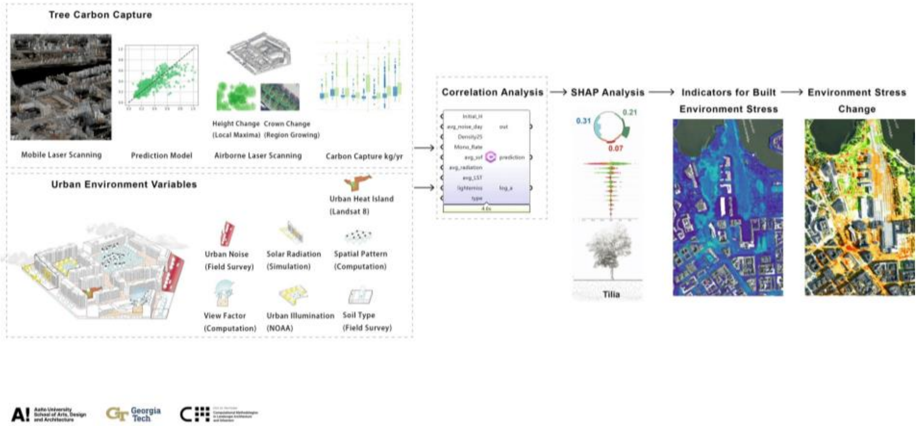
成果案例 Case Workshops



Shi Y, Hong TK (2026) Benchmarking Pix2Pix: Shape Grammar as a Deterministic Standard for Model Limits. The Association for Computer-Aided Architectural Design Research in Asia (CAADRIA) 2026.



Zhu, C., Susskind, J., Giampieri, M., O'Neil, H. B., & Berger, A. M. (2023). Optimizing sustainable suburban expansion with autonomous mobility through a parametric design framework. Land, 12(9), 1786.



Yao, C., Gao, S., Fricker, P., & Shi, Y. (In Progress). Improving Urban Tree Growth Conditions for Carbon Sequestration: A Spatial Machine Learning Analysis of Built-Environment Effects.

Workflow Prototype

1 涵盖模型训练，变量解释，性能模拟以及设计生成的紧凑可验证的工作流。

An end-to-end validated pipeline from design generation through performance evaluation and interpretation.

Grasshopper Plugin

2 将代理模型和 XAI 集成的轻量级可分发的 Rhino 工具。

A lightweight, distributable Rhino tool integrating surrogate models and XAI.

Conference Publication

3 展示方法与初步研究发现的会议论文。

A conference paper presenting the method and initial findings.

27.06.2026

Afternoon

2:00 - 2:30 **Overall Scheduling Introduction**

2:30 - 3:30 **Participants‘ introduction** (1-2min per person),

3:45 - 4:30 **Methodology Lecture – Part 1**
(What is surrogate model, How to extract feature, How to train surrogate model, Chaowen)

4:30 - 5:00 **Track Division** (Arch, Urban, Landscape tracks, Chaowen)

Evening **Environment Setup**(Rhino8, Colab/Jupyter, necessary Libraries, Yichao)

Thinking Ideas

28.06.2026

Morning

9:30 -10:30 **Methodology Lecture – Part 2**
(Encapsulate machine learning models: why encapsulate, and how to encapsulate them into GH, Yichao)

10:45 -11:45 **Methodology Lecture – Part 3**
(Explainable AI, generative design, nonlinear optimization, and machine learning-based optimization scheme selection., Chenhao)

Afternoon

2:00 - 3:00	**Case&Demo** (Chaowen, Landscape/Planning Topic)
3:00 – 4:00	**Prof. Lecture** (Digital Computation, Pia)
4:00 – 5:00	**Case&Demo** (Chenhao, Urban Design Topic)
5:00 - 6:00	**Topic and Grouping Discussion**
Evening	**Deepen Proposal**

29.06.2026

Morning

9:00 - 9:45	**Case (**Yichao, Architecture Topic)
10:00 - 10:45	**Demo** (Encapsulation demonstration, Yichao)
11:00 - 12:00	**Team Confirmation**

Afternoon

Office Hour (Hybrid)

Evening

Office Hour (Yichao, online)

30.06.2026

Morning

Midterm presentation (Chaowen, Chenhao, Yichao)

Afternoon

Office Hour (Hybrid)

Evening

Office Hour (Yichao, online)

01.07.2026

Morning

Track + Team meeting (Flexible)

Afternoon

Office Hour (Hybrid)

Evening

Office Hour (Yichao, online)

02.07.2026

Morning

Track + Team meeting (Flexible)

Afternoon

Office Hour (Hybrid)

Evening

Prepare Layout & Slides

03.07.2026

Morning

Track + Team meeting (Flexible)

Afternoon

2:00 - 5:00 Final Review